Trade Restriction, Industrial Policy, and Comparative Advantage

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May 7, 2026

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1. Introduction

In today's global economy, we are no longer operating under the traditional rules of free trade, but instead in a period of increasing trade tension and uncertainty. Rising geopolitical competition, especially between the United States and China, has pushed countries to rethink how they engage in international trade. Rather than being simply focused on efficiency and comparative advantage, policy decisions are becoming increasingly influenced by concerns over supply chain dependence, domestic production capacity, and economic resilience. From tariffs on Chinese goods to subsidies for domestic manufacturing, governments are actively intervening in markets in ways that challenge the core assumptions of classical trade theory.

One of the clearest examples of this can be seen in industries such as drone manufacturing. Over the past decade, Chinese firm DJI has come to dominate the global drone market through large-scale production, vertically integrated supply chains, and significantly lower prices (e.g., Congressional Research Service, 2023; Politico, 2025). By contrast, U.S. firms have struggled to compete in commercial markets, despite maintaining strengths in software and high-end innovation (Congressional Research Service, 2023). In response, U.S. policymakers have introduced tariffs, restrictions on foreign technology, and industrial policy initiatives aimed at rebuilding domestic production capacity. Similar patterns can be observed in other sectors, such as semiconductors, where government intervention has increased through large-scale subsidy programs such as the CHIPS and Science Act. These projects have faced delays in production, rising costs, and labor shortages, highlighting the difficulty of rapidly rebuilding domestic capacity.

These developments raise an important economic question: Do trade restrictions and industrial policy effectively reshape comparative advantage, or do they primarily create higher costs and inefficiencies in the economy? This question is central to understanding the current direction of global trade policy. Traditional trade theory, particularly the Ricardian model, suggests that countries benefit from specializing according to their comparative advantage, leading to lower prices and higher overall welfare. However, recent policies appear to challenge this framework by attempting to shift production away from low-cost foreign producers and toward domestic industries, even when those industries are less efficient.

This paper examines this question by focusing on the interaction between trade restrictions, industrial policy, and supply chain restructuring. Using the drone industry as a primary case study, along with broader U.S.–China trade data and price trends, the analysis explores how these policies affect production, trade flows, and consumer welfare. Drawing on concepts from international trade theory, including comparative advantage, economies of scale, and welfare analysis, the paper evaluates whether policy interventions can meaningfully alter economic outcomes or whether they primarily result in higher costs and reduced efficiency.

Understanding this issue is important not only from a theoretical perspective, but also for real-world policy decisions. As governments continue to prioritize domestic production and reduce reliance on global supply chains, the trade-offs between efficiency, cost, and economic resilience become increasingly relevant. By examining these trade-offs, this paper aims to provide a clearer understanding of whether current trade policies represent a sustainable shift in comparative advantage or a costly deviation from the principles of efficient global trade.

2. Literature Review

The role that trade restrictions play in determining comparative advantage highlights an intersection in economic literature between classical trade theory and modern empirical studies on trade conflicts. While the foundational theory of comparative advantage, developed by David Ricardo, demonstrates that specialization leads to lower prices, higher total output, and overall welfare gains, more recent empirical work highlights the strategic motivations for intervention and the economic costs that accompany it.

Paul Krugman (1980) introduces the concept that economies of scale and network effects can shape comparative advantage, further complicating Ricardo's model. In industries characterized by increasing returns, early leaders can gain the ability to dominate global markets. This framework helps explain the dominance of DJI in drone manufacturing by gaining cost advantages over time through large-scale production and integrated supply chains. In this context, industrial policy could theoretically shift comparative advantage by helping domestic firms reach efficient scale, although this process is slow and uncertain.

Separately, in *The Globalization Paradox*, Dani Rodrik (2011) emphasizes the political and strategic motivations behind trade policy, arguing that deep international trade liberalization can conflict with national sovereignty and domestic priorities. This suggests that trade restrictions and industrial policy are not merely economic distortions but tools for maintaining political control and addressing domestic concerns such as employment, technological independence, and economic security. Similarly, Farrell and Newman (2019) introduce the concept of "weaponized interdependence," arguing that countries controlling key nodes in global

networks can gain strategic power. This helps explain why sectors such as semiconductors and drones have become focal points of trade restrictions.

Contrary to these justifications, empirical research from the recent U.S.–China trade war provides strong evidence that tariffs generate significant economic costs. Fajgelbaum et al. (2020) find that the costs of U.S. tariffs were almost entirely passed on to domestic consumers, leading to substantial welfare losses. Their results show that real income declined due to higher prices and reduced trade. Similarly, Amiti, Redding, and Weinstein (2019) find that firms exposed to tariffs reduced investment and experienced increased input costs, suggesting that protectionist policies can weaken rather than strengthen domestic production.

Additional research further reinforces this point. Autor, Dorn, and Hanson (2016) show how trade shocks can reshape labor markets but do not necessarily lead to efficient reallocation without significant adjustment costs. Baldwin (2016) highlights the role of global value chains, emphasizing that modern production is deeply interconnected, making trade disruptions particularly costly. At the same time, Grossman, Helpman, and Lhuillier (2021) develop models of supply chain resilience, showing that while reshoring policies can reduce risk, they often increase costs and reduce efficiency.

When brought together, the literature presents a nuanced picture. While theoretical and policy-oriented research suggests that governments may pursue trade restrictions for strategic reasons, both classical models and empirical evidence consistently show that such policies impose welfare losses through higher prices, reduced investment, and inefficient resource allocation.

3. Economic analysis

3.1 Subsidy and Industrial Policy

In recent years, U.S. trade and industrial policy has increasingly shifted focus toward national security and domestic production capacity, particularly in strategic sectors such as drone technology. Rather than relying purely on global comparative advantage, policymakers have begun actively intervening to reshape market outcomes through trade restrictions, subsidies, and government procurement programs. The drone industry provides a clear example of this policy-induced attempt to transform comparative advantage.

The Chinese firm DJI has achieved near-global dominance through mass production, vertically integrated supply chains, and cost efficiency. China dominates global drone production, manufacturing millions of units annually, with some estimates reaching tens of millions, and controls roughly 70–80% of the commercial drone market. In contrast, the United States produces significantly fewer drones, often under 100,000 to 200,000 units per year, and remains heavily reliant on foreign supply chains for key components such as batteries, sensors, and cameras (Congressional Research Service, 2023).

This imbalance is not simply a matter of firm performance, but reflects deeper structural differences in how production is organized across countries. China's advantage comes from economies of scale and a highly coordinated manufacturing ecosystem, where production, inputs, and assembly are tightly integrated within regional clusters such as Shenzhen. This allows firms like DJI to reduce costs and rapidly scale output. As a result, Chinese drones are significantly cheaper: consumer models range from $300–$500, while enterprise models are typically priced between $10,000 and $13,000. By comparison, the U.S.-produced drones such as the Skydio

X10 or Freefly Astro Max can cost between $15,000 and $26,000. These persistent price differences highlight the role of productivity and scale in determining comparative advantage.

From a Ricardian model perspective, China holds a comparative advantage in large-scale drone manufacturing due to lower unit production costs, while the United States specializes in higher value-added activities such as software development, system integration, and defense-related innovation. As shown in the left panel of Figure 1, this is reflected in the shape of each country's production possibility frontier: China's PPF is relatively flat along the drone axis, indicating a low opportunity cost of drone production, while the U.S. PPF is steep, reflecting high opportunity cost in manufacturing but strength in higher-value sectors. Each country specializes where the world price line is tangent to its frontier. This distinction is critical because it implies that comparative advantage is rooted in productivity differences that cannot be easily altered in the short run.

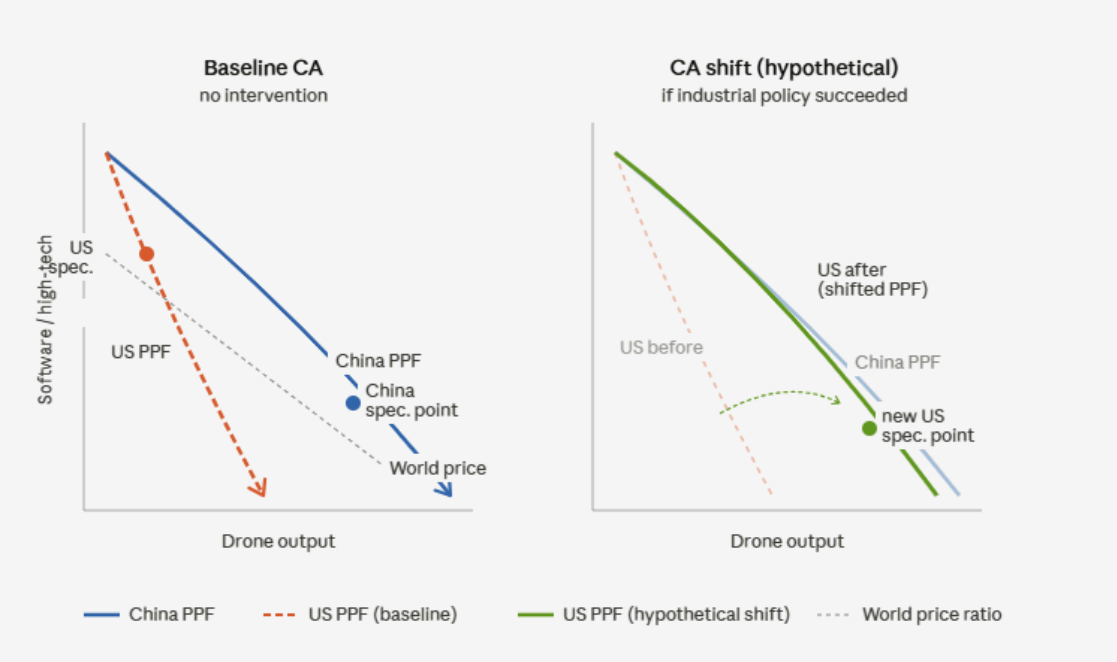


Figure 1: US PPF Comparison

The right panel of Figure 1 illustrates what a successful policy intervention would need to accomplish. For industrial policy to genuinely shift comparative advantage, U.S. productivity in drone manufacturing would need to improve enough to rotate the PPF outward toward the drone axis, lowering the opportunity cost of production and moving the U.S. specialization point rightward toward China's. This is the theoretical promise of the infant industry argument, or idea that temporary protection can allow new industries to achieve scale and productivity gains, enabling them to become internationally competitive.

Recognizing this gap, U.S. policymakers have attempted to intervene directly in market outcomes. Programs such as the Department of Defense's Blue UAS initiative and procurement strategies aimed at expanding domestic drone production seek to reduce reliance on foreign suppliers. Similarly, broader industrial policies, including subsidies and tax incentives, aim to encourage domestic manufacturing, with the goal of achieving exactly the kind of PPF rotation shown in the right panel of Figure 1.

However, this transition is not costless. Restricting access to lower-cost imports forces consumers, firms, and government agencies to substitute toward higher-cost domestic alternatives. In Ricardian terms, this is reflected by moving up the existing steep U.S. frontier rather than expanding it. In the short run, this does not eliminate China's cost advantage but instead raises prices and reduces overall efficiency.

3.2 Supply Chain Reorganization

The shift in industrial policy observed in the drone industry reflects a broader restructuring of global supply chains. For decades, the U.S.–China trade relationship has been characterized by a persistent goods trade deficit. While often framed as a policy failure, this deficit also reflects

structural features of the U.S. economy, including high consumption, strong demand for manufactured goods, and specialization in services rather than large-scale manufacturing. As a result, the United States has relied heavily on imports, with China accounting for approximately 15–18% of total U.S. imports in recent years, particularly in manufacturing-intensive sectors such as electronics, machinery, and consumer goods (U.S. Census Bureau, 2024; Congressional Research Service, 2023). This pattern is consistent with the predictions of comparative advantage, where countries specialize in goods they can produce relatively more efficiently.



Figure 2: US Share of Imports from China (US Census Bureau)

Recent trade policies have attempted to alter this pattern by restricting imports and encouraging domestic production. Figure 2 shows that imports from China have declined following the introduction of tariffs and trade barriers. However, U.S. exports have also declined during the same period (Amiti, Redding, & Weinstein, 2019; Fajgelbaum et al., 2020). Rather than strengthening domestic production, trade restrictions are associated with a reduction in overall trade activity. In other words, the adjustment mechanism is not “replace imports with domestic output,” but rather “reduce both imports and exports.” This suggests that trade policy is

disrupting existing economic relationships rather than generating new productive capacity. From a trade theory perspective, this outcome reflects a contraction in gains from trade rather than a reallocation toward a new comparative advantage.

At a deeper level, this outcome reflects the complexity of modern supply chains. Production today is not confined within national borders but is distributed across multiple countries, each specializing in different stages of production (Baldwin, 2016; Antràs, 2020). For example, drones rely on batteries, sensors, semiconductors, and software developed across different regions. This fragmented production structure allows firms to minimize costs and maximize efficiency by exploiting differences in factor endowments, labor costs, and technological capabilities. However, it also creates interdependence, where disruptions in one part of the supply chain can have cascading effects across the entire production network.

As a result, supply chains are no longer organized purely by cost efficiency, but increasingly by security, political alignment, and strategic considerations. Governments are attempting to reshore or "friend-shore" production to reduce reliance on geopolitical rivals. For example, the United States has encouraged semiconductor production through domestic investment incentives such as the CHIPS and Science Act, while also coordinating with allied countries to diversify supply chains (Antràs, 2020). These policies reflect a shift away from efficiency-driven globalization toward resilience-oriented production strategies.

Importantly, these transitions introduce significant constraints. Building domestic capacity requires time, skilled labor, infrastructure, and coordination across industries. Empirical evidence suggests that reshoring production is both costly and slow, as firms must rebuild supplier networks and invest in new production facilities (McKinsey Global Institute, 2020; IMF,

2023). Even with strong policy support, projects such as semiconductor manufacturing plants have faced delays, cost overruns, and workforce shortages, highlighting the difficulty of replicating established global production ecosystems.

These structural constraints are important because they shape the ultimate economic outcomes of trade policy. If supply chains cannot adjust quickly, then the immediate impact of trade restrictions will not appear as increased domestic production, but rather as higher prices and reduced efficiency across the economy. This reinforces a key argument of this paper: while trade restrictions and industrial policy aim to reshape economic structure, their short-run effects are more consistent with reduced trade, increased costs, and disruption of existing comparative advantage rather than the creation of a new one.

3.3 Welfare Trade-Off

The economic consequences of these policy changes become most visible when examining price data. While policymakers aim to increase domestic production and enhance economic security, the short-run effects are primarily felt through changes in prices and consumer welfare.

As shown in Figure 3, U.S. import prices begin to rise above their projected trend after 2025. The data are based on U.S. import price indices (Bureau of Labor Statistics), normalized to an index where December 2024 = 100. The solid line represents actual observed import prices, while the dashed line represents a simple linear projection based on pre-2025 trends. Prior to this period, prices followed a relatively stable trajectory. However, following the introduction of tariffs and trade restrictions, prices deviate upward from the projected trend. This indicates that limiting access to lower-cost imports forces the domestic economy to substitute toward

higher-cost alternatives. The upward shift in prices reflects a loss of efficiency consistent with reduced gains from trade when global specialization is disrupted.

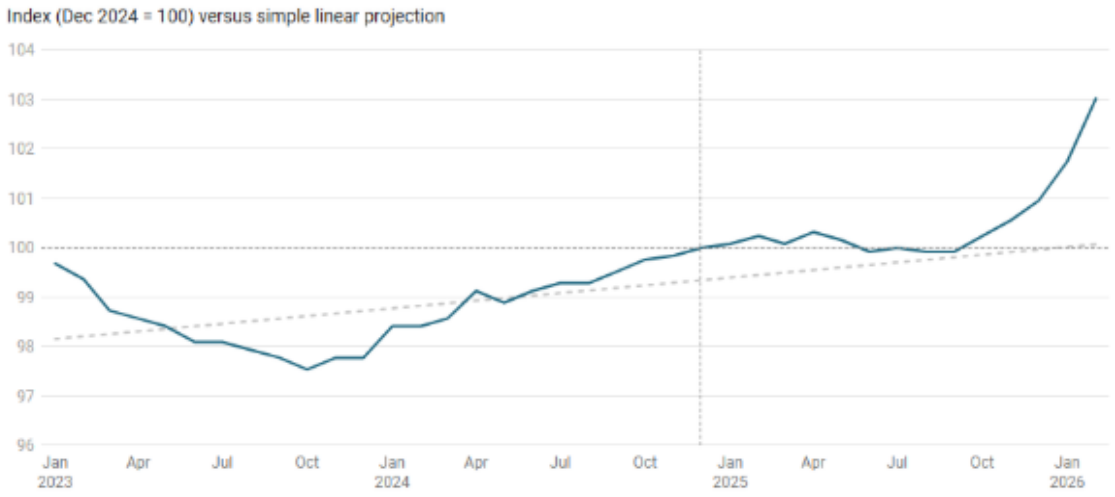


Figure 3: Non-Petroleum Import Prices (Bureau of Labor Statistics)

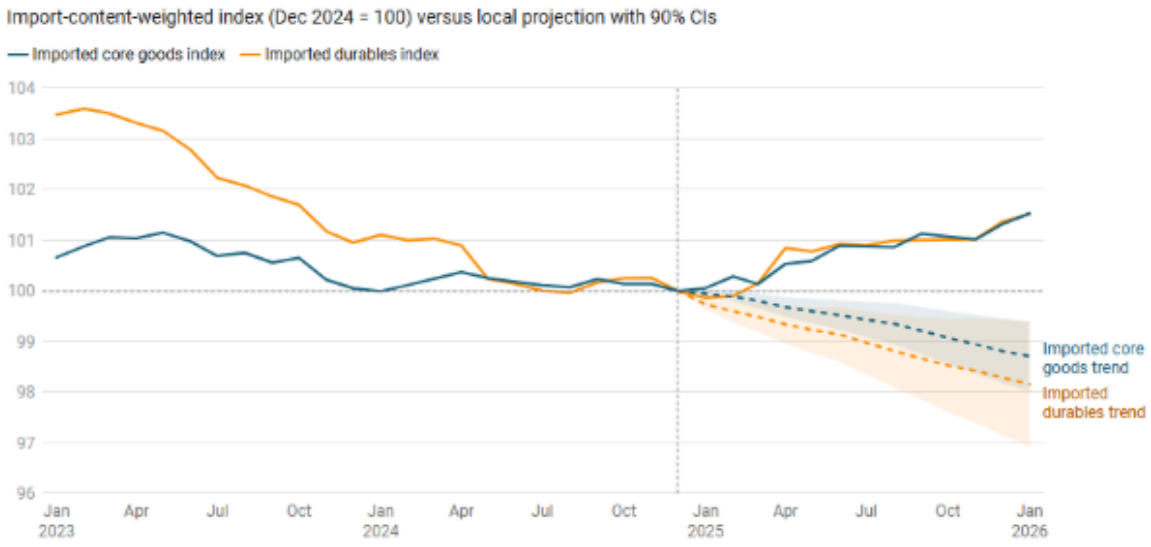


Figure 4: Imported PCE Core Goods & Durables Prices (BEA)

This pattern becomes more pronounced in Figure 4. These data are drawn from the Bureau of Economic Analysis (BEA) Personal Consumption Expenditures (PCE) price indices, adjusted for import content and again normalized to December 2024 = 100. The solid lines represent actual observed price indices for imported core goods and durable goods, while the dashed lines represent projected trends based on pre-policy data. The shaded regions reflect confidence intervals around the projections. After 2025, both core goods and durable goods diverge from their expected paths, with durable goods experiencing particularly strong increases. This is consistent with the fact that durable goods such as electronics, machinery, and vehicles are more deeply embedded in global supply chains and therefore more sensitive to trade disruptions (Baldwin, 2016; Antràs, 2020). The divergence from trend suggests that these price increases are not driven by normal demand fluctuations, but by structural changes in trade policy affecting input costs and production networks.

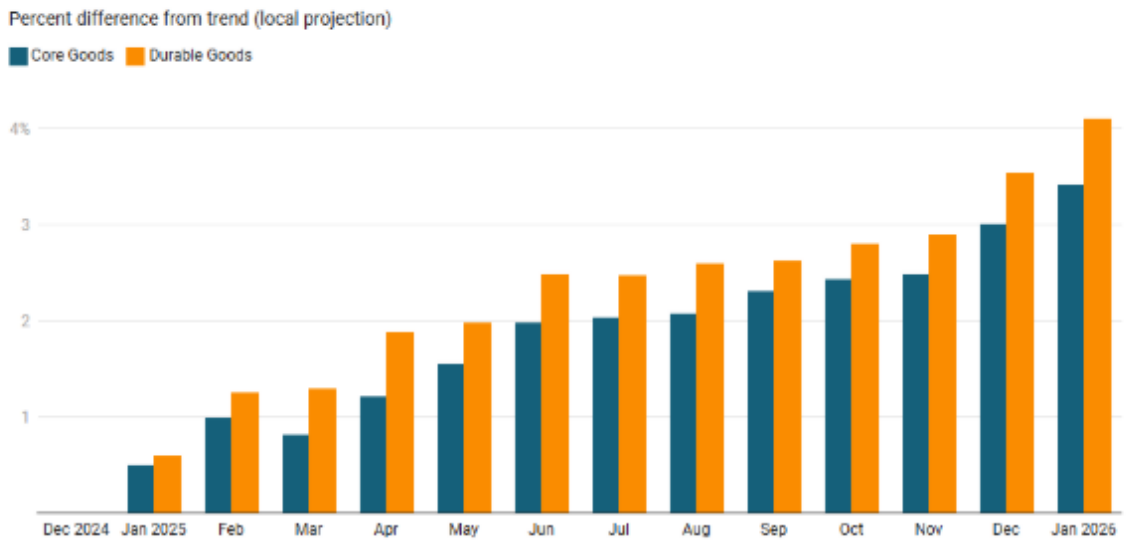


Figure 5: PCE Goods Prices: Deviation from Trend (BEA)

The magnitude of these effects is further illustrated in Figure 5. This figure measures the percentage difference between actual prices and their projected baseline levels. Positive values indicate that prices are higher than expected under normal conditions. The consistent upward deviations across both core goods and durable goods confirm a well-established result in trade economics: tariffs act as a tax on imports, raising domestic prices. Importantly, this is not a theoretical prediction being tested here, but an empirical outcome observed in the data. The burden of tariffs is largely passed through to domestic consumers in the form of higher prices. The persistence of these deviations suggests that the impact of tariffs is not temporary, but reflects a sustained increase in price levels associated with reduced access to low-cost global production.

At the same time, the expected benefits of protection are offset by the response of trading partners. Retaliation introduces additional economic costs, particularly for export-oriented sectors. China imposed tariffs on key U.S. exports such as soybeans, reducing their competitiveness relative to alternative suppliers. This led to a significant decline in U.S. export share in global markets. In response, the U.S. government provided billions of dollars in subsidies to affected farmers, effectively redistributing the cost of trade policy from exporters to taxpayers (Fajgelbaum et al., 2020). This cycle of tariffs, retaliation, and subsidies demonstrates that the economic burden of protectionism does not disappear; rather, it is reallocated across different groups within the domestic economy.

Finally, the effectiveness of these policies must be evaluated in terms of their impact on domestic production. As shown in Figure 6, U.S. manufacturing output measured using Federal Reserve industrial production indices does not exhibit a strong or sustained increase following the implementation of trade restrictions. The solid line represents actual industrial production,

while the dashed line represents a projected trend based on pre-2025 growth patterns. Although there are short-term fluctuations, the overall level of production remains relatively flat compared to its expected trajectory. This suggests that reducing imports does not automatically translate into increased domestic production. Instead, it highlights the difficulty of rebuilding industrial capacity in the presence of structural constraints such as productivity differences, economies of scale, and supply chain dependence.



Figure 6: Manufacturing Industrial Production (BEA)

4. Conclusion

This paper examines whether trade restrictions and industrial policy effectively reshape comparative advantage, or whether they primarily generate higher costs and inefficiencies. The evidence presented suggests that while these policies can influence trade patterns and reduce dependence on foreign producers, their immediate economic effects are largely characterized by increased costs and limited improvements in domestic production capacity.

The case of the drone industry illustrates how comparative advantage is deeply rooted in productivity, scale, and supply chain integration. China's dominance in large-scale manufacturing allows firms like DJI to produce at significantly lower costs, while the United States remains more competitive in high-value innovation and specialized technologies. Attempts to shift this advantage through tariffs and industrial policy do not eliminate these underlying differences, but instead raise prices and reduce the gains from trade. Empirical evidence supports this outcome, showing that import prices and consumer good prices rise above their projected trends following the implementation of trade restrictions, indicating that a significant portion of the burden is passed on to domestic consumers.

At the same time, the broader restructuring of supply chains highlights the limits of these policies. While reducing imports from China has narrowed the bilateral trade deficit, it has also reduced exports and overall trade activity. This suggests that trade restrictions do not necessarily strengthen domestic industries, but may instead disrupt existing economic relationships. Furthermore, domestic manufacturing output has not experienced a sustained increase, meaning that reshoring efforts have not yet translated into meaningful production gains.

The role of retaliation further reinforces this conclusion. Tariffs imposed by trading partners reduce the competitiveness of U.S. exports, particularly in sectors such as agriculture, leading to significant losses for domestic producers. Government responses in the form of subsidies demonstrate that these policies do not eliminate economic costs, but rather redistribute them across consumers, producers, and taxpayers.

Taken together, these findings suggest that trade restrictions and industrial policy have limited effectiveness in reshaping comparative advantage in the short run. While they can alter

incentives and shift production patterns at the margin, they cannot easily overcome structural constraints such as economies of scale, established supply chains, and productivity differences. As a result, these policies tend to generate higher prices, reduced efficiency, and increased economic distortion.

Ultimately, this analysis suggests that trade restrictions function less as a tool for reshaping comparative advantage and more as a mechanism for reallocating costs within the domestic economy. While arguments for economic resilience and supply chain security may justify some level of intervention, the evidence indicates that these benefits are uncertain and slow to materialize. In contrast, the costs in the form of higher prices, reduced trade, and government expenditures are immediate and measurable. Therefore, while trade restrictions and industrial policy can influence the direction of economic activity, they primarily create higher costs and inefficiencies rather than fundamentally reshaping comparative advantage in a meaningful and sustainable way.

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